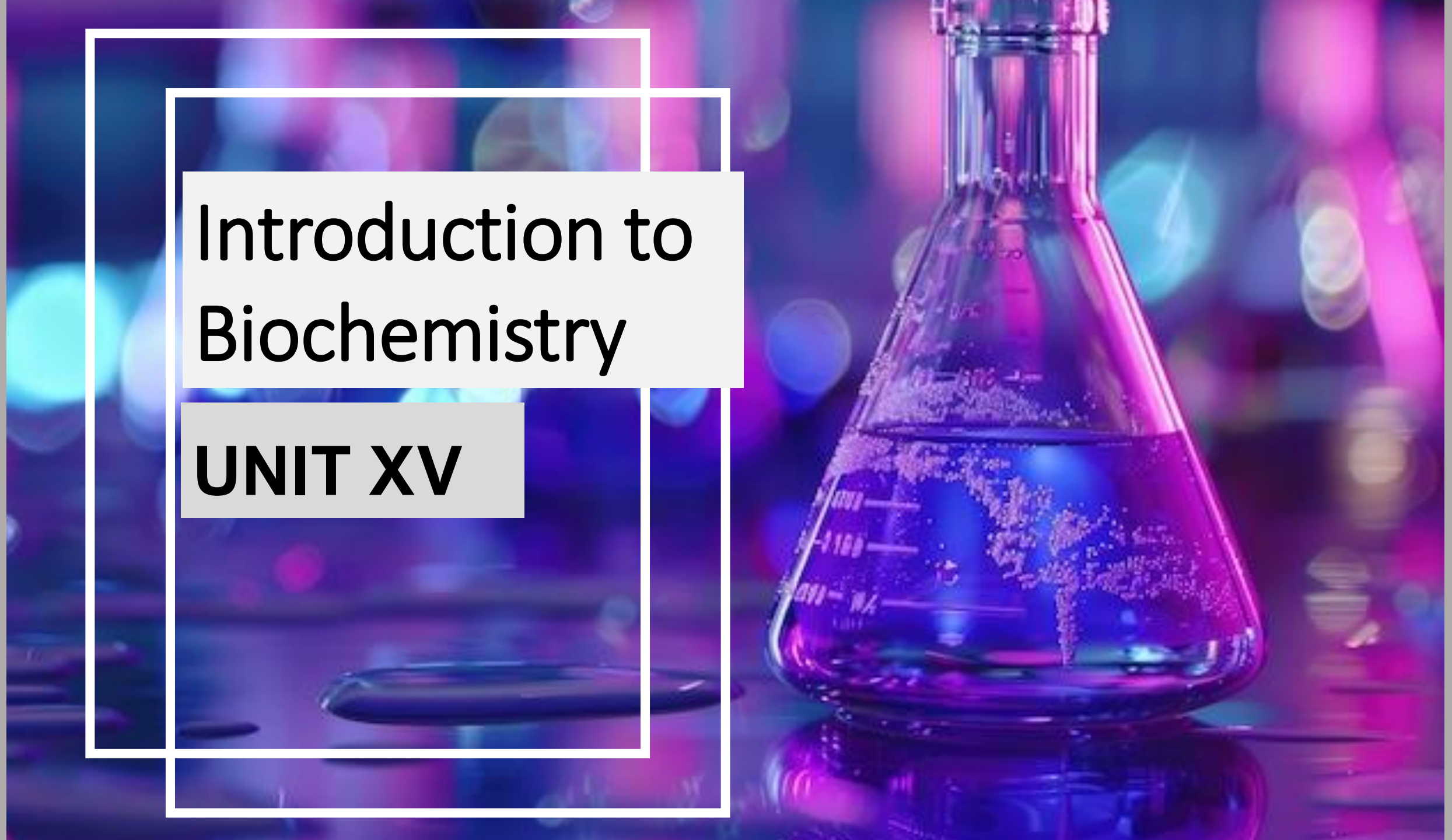


Introduction to Biochemistry

UNIT XV

Learning Outcomes

At the end of each topic, the students should be able to:

- ❑ Appreciate the scope of biochemistry and its central role in life sciences, and that biochemistry and medicine are intimately related disciplines
- ❑ Appreciate that biochemistry integrates knowledge of the chemical processes in living cells with strategies to maintain health and enhance our understanding of the origins of life on earth
- ❑ Provide an overview of the biomolecules and to define biochemistry and how is it different from other natural sciences
- ❑ Review on the basic concepts of organic chemistry to understand the properties of biomolecules

Biochemistry - An Overview

- Study of chemical substances found in living organisms and the chemical interactions of these substances with each other
- Compounds, chemical reactions, and molecular interactions involved in production maintenance and reproduction of living organisms
- How cells manufacture the molecules needed for life and how the chemical reactions by which life is maintained occur



Eating provides energy for activities like flight: A hummingbird needs energy to maintain prolonged periods of flight. The hummingbird obtains its energy from taking in food and transforming the nutrients into energy through a series of biochemical reactions. The flight muscles in birds are extremely efficient in energy production.

https://courses.lumenlearning.com/boundless-biology/chapter/energy-and-metabolism/

- All living things make use of same type of biomolecules and use energy. Hence studied through methods of chemistry and physics
- Fundamental similarity of cells and speculating on origin of life
- Cells and biomolecules arisen from very simple molecules such as H_2O, CH_4, NH_3, N_2 and H_2
- Field of biochemistry draws many disciplines
- Answer questions on molecular nature of life

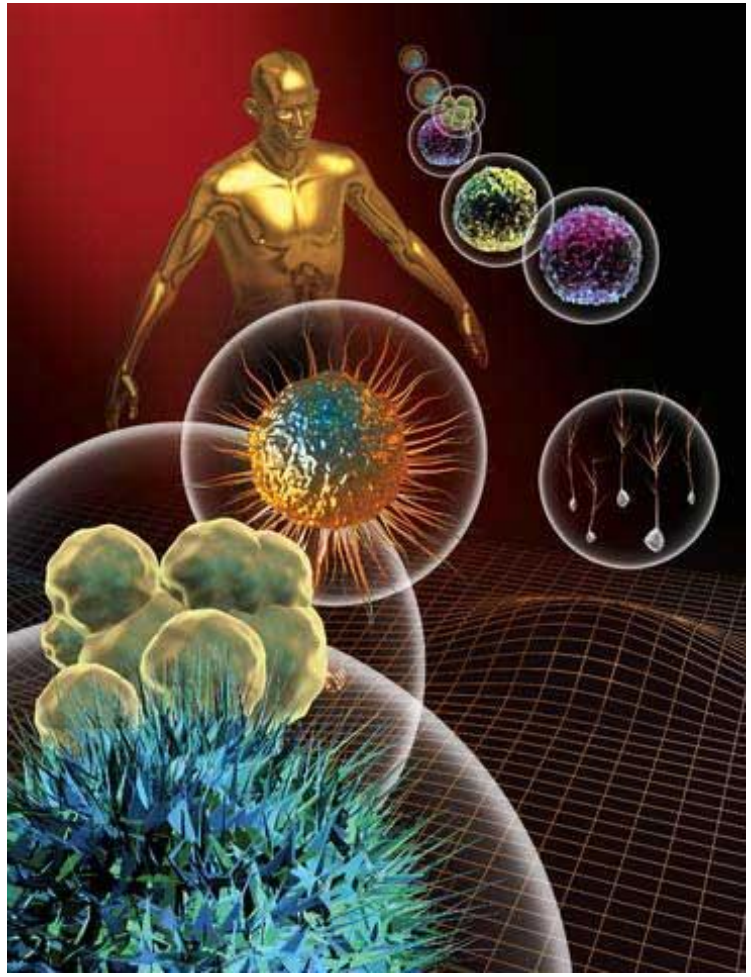
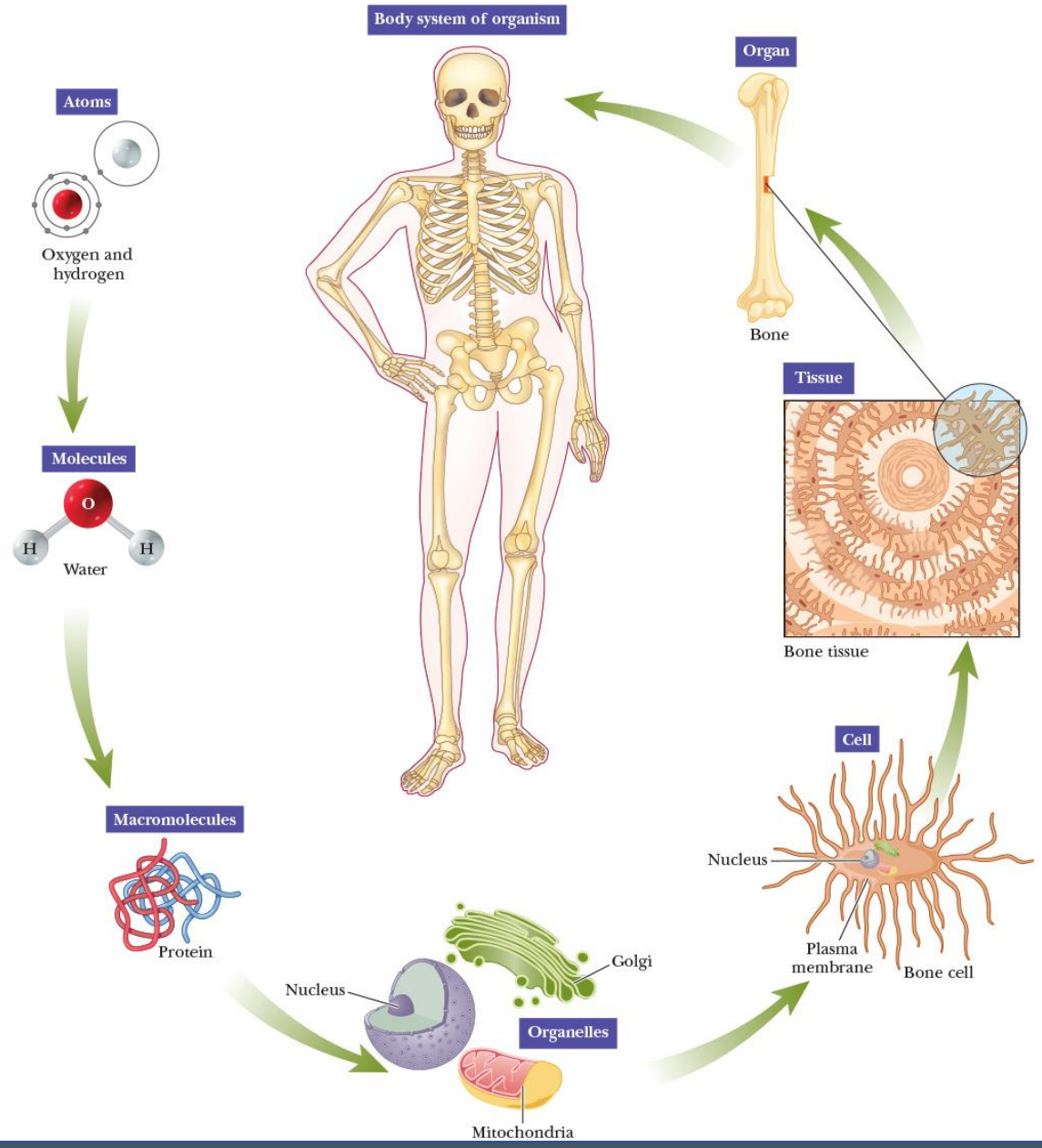


Figure 1.1 - Levels of Structural Organization in the Human Body



Chemical Foundations of Biochemistry

Biochemistry seeks to describe the structure, organization, and function of living organisms in molecular terms

In order to understand the life on the molecular level, you must first have a:

- Knowledge of the chemical structures of the biological molecules
- Understanding of the biological function of the molecules
- Understanding of bioenergetics - the study of energy flow in cells

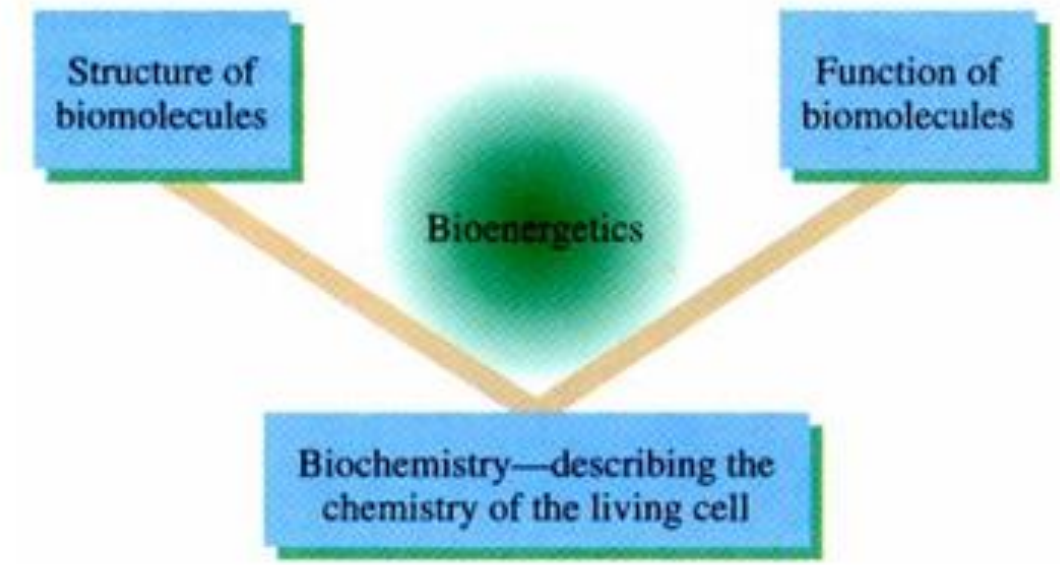


Figure 1.2 Molecular scope of biochemistry

Chemical Foundations of Biochemistry

Organic Chemistry: study of compounds of carbon

- Cellular apparatus of living organisms made up of carbon
- **Biomolecules** are part of organic chemistry
- Reactions of biomolecules can be described by organic chemistry methods
- **Functional Groups:** atom/group of atoms that show characteristic physical and chemical properties

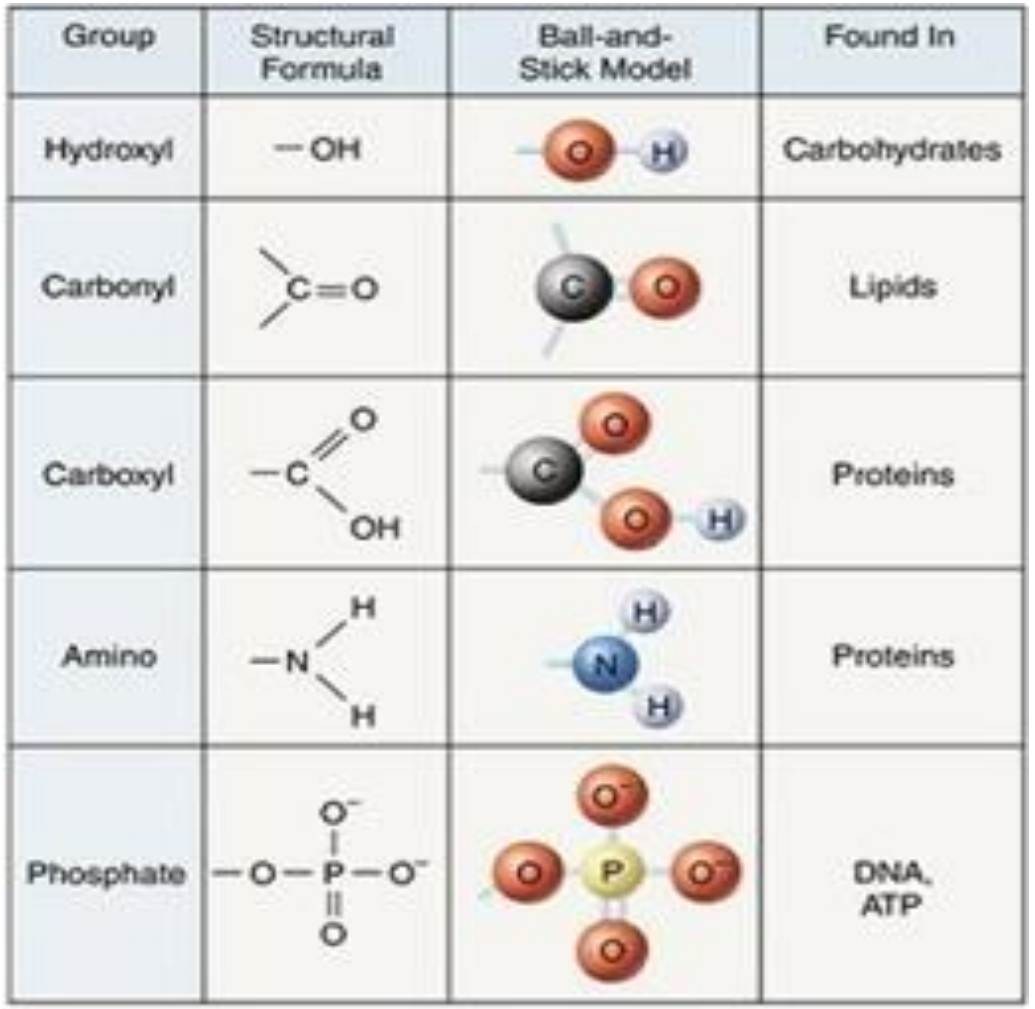


Figure 1.3 Functional groups common in biochemistry © Weebly.com

Chemical Foundations of Biochemistry

Biomolecules: cells include large molecules such as proteins, nucleic acids, polysaccharides and lipids

- These biomolecules are polymers (Greek: ***poly*** + ***meros,*** many + parts)
- Derived from monomers (Greek: ***mono*** + ***meros,*** single + parts)

Biological Micromolecules	Biological Macromolecules
Amino Acid	Protein
Sugars	Carbohydrates
Nucleotides	Nucleic acid
Lipids	

- ***Proteins and Nucleic Acids play a role in life processes.***

Chemical Foundations of Biochemistry

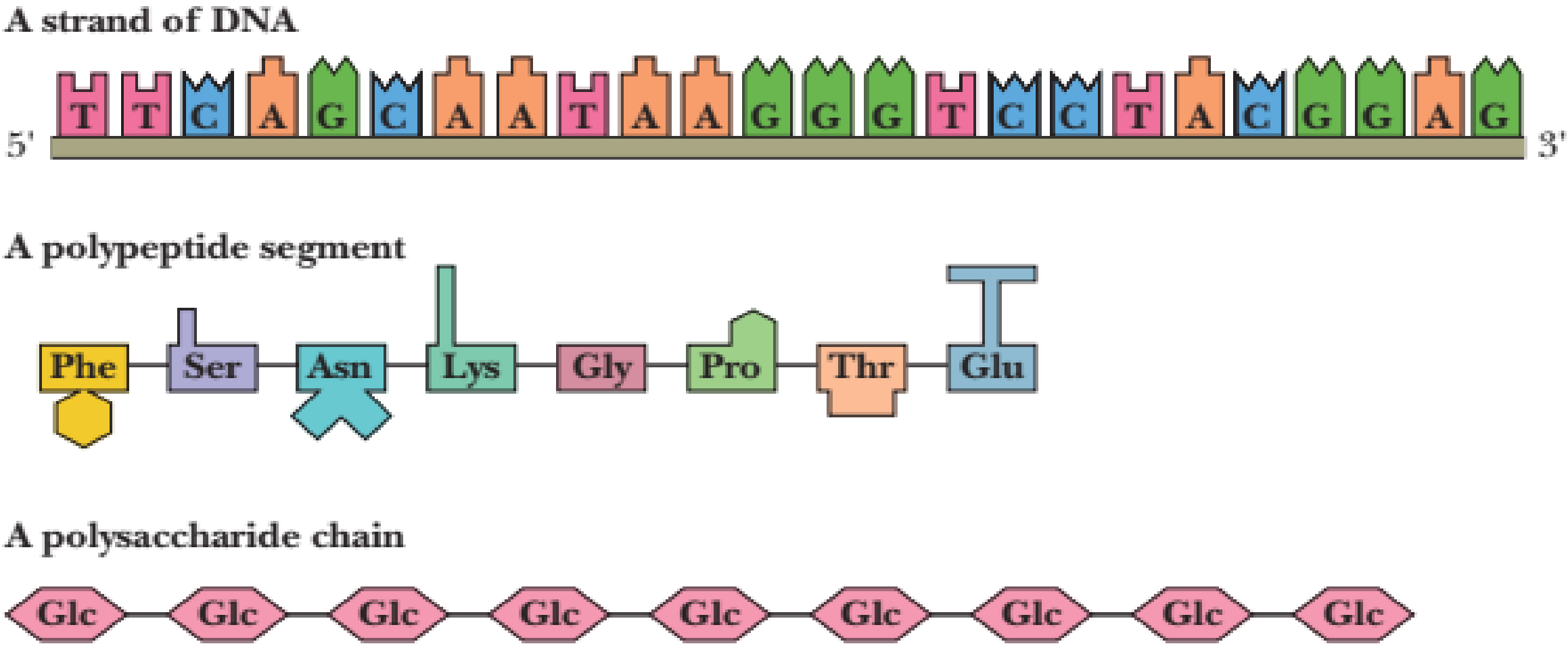
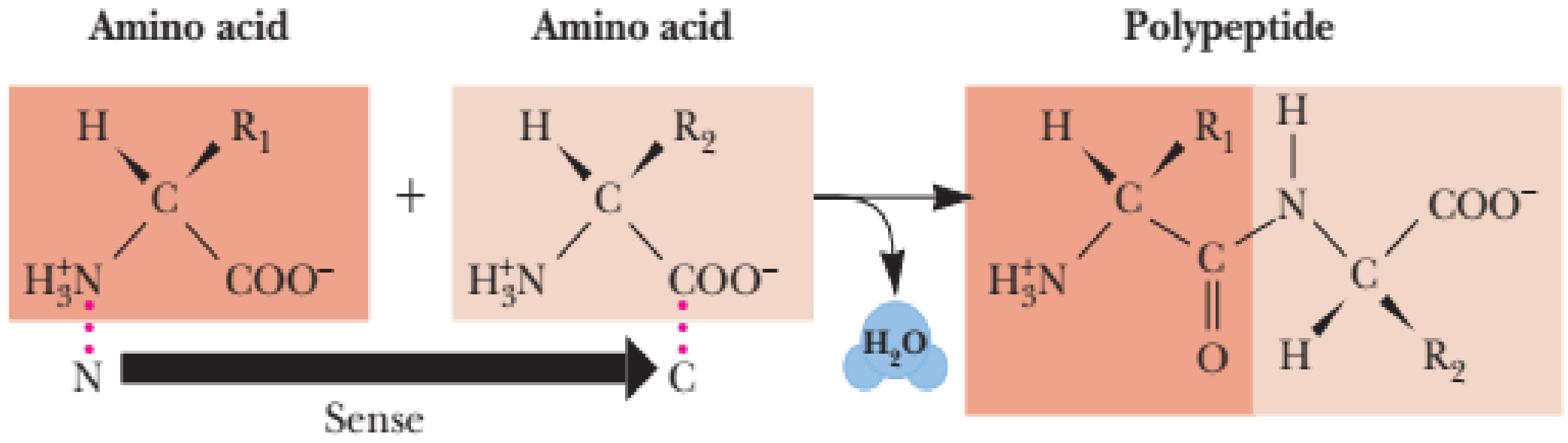


Figure 1.4 Informational Macromolecules

© Cengagelearning. All rights reserved. (Campbell & Farrell, 2007)

Chemical Foundations of Biochemistry



A Amino acids build proteins by connecting the carboxyl group of one amino acid with the amino group of the next amino acid.

Figure 1.5 Directionality in Polypeptides

© Cengagelearning. All rights reserved. (Campbell & Farrell, 2007)

Chemical Foundations of Biochemistry

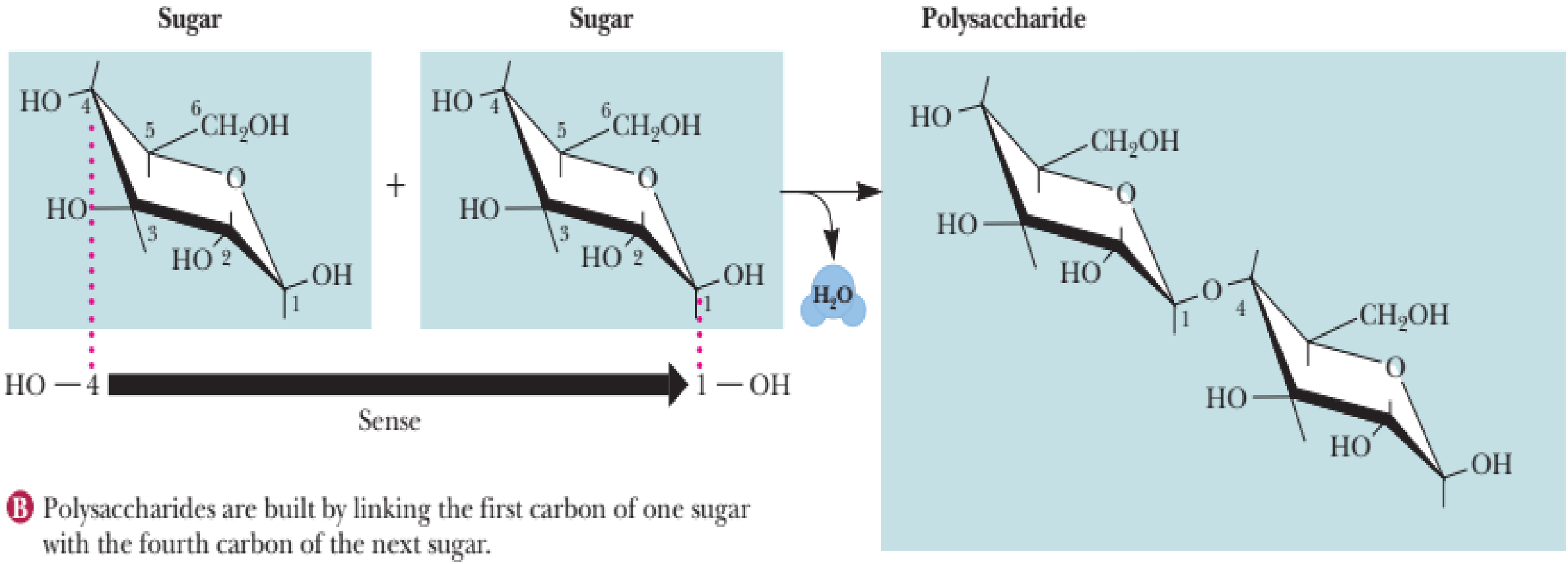
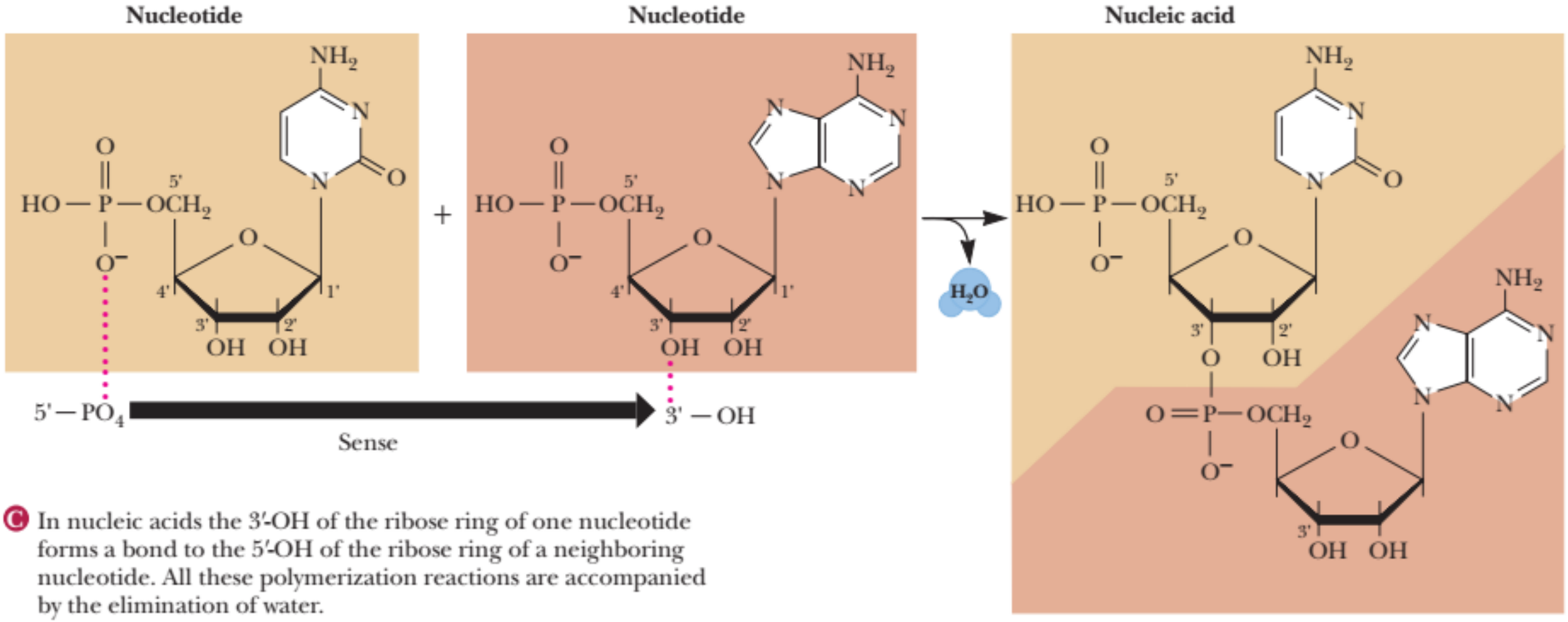


Figure 1.6 Directionality in Polysaccharides

© Cengagelearning. All rights reserved. (Campbell & Farrell, 2007)

Chemical Foundations of Biochemistry



C In nucleic acids the 3′-OH of the ribose ring of one nucleotide forms a bond to the 5′-OH of the ribose ring of a neighboring nucleotide. All these polymerization reactions are accompanied by the elimination of water.

Figure 1.7 Directionality in Nucleic Acid

© Cengagelearning. All rights reserved. (Campbell & Farrell, 2007)

EXERCISE

Biochemistry, the study of the chemical substances of living organisms, includes the study of:

a.carbohydrates, lipids, proteins, and salts.

b.carbohydrates, lipids, proteins, and organic salts.

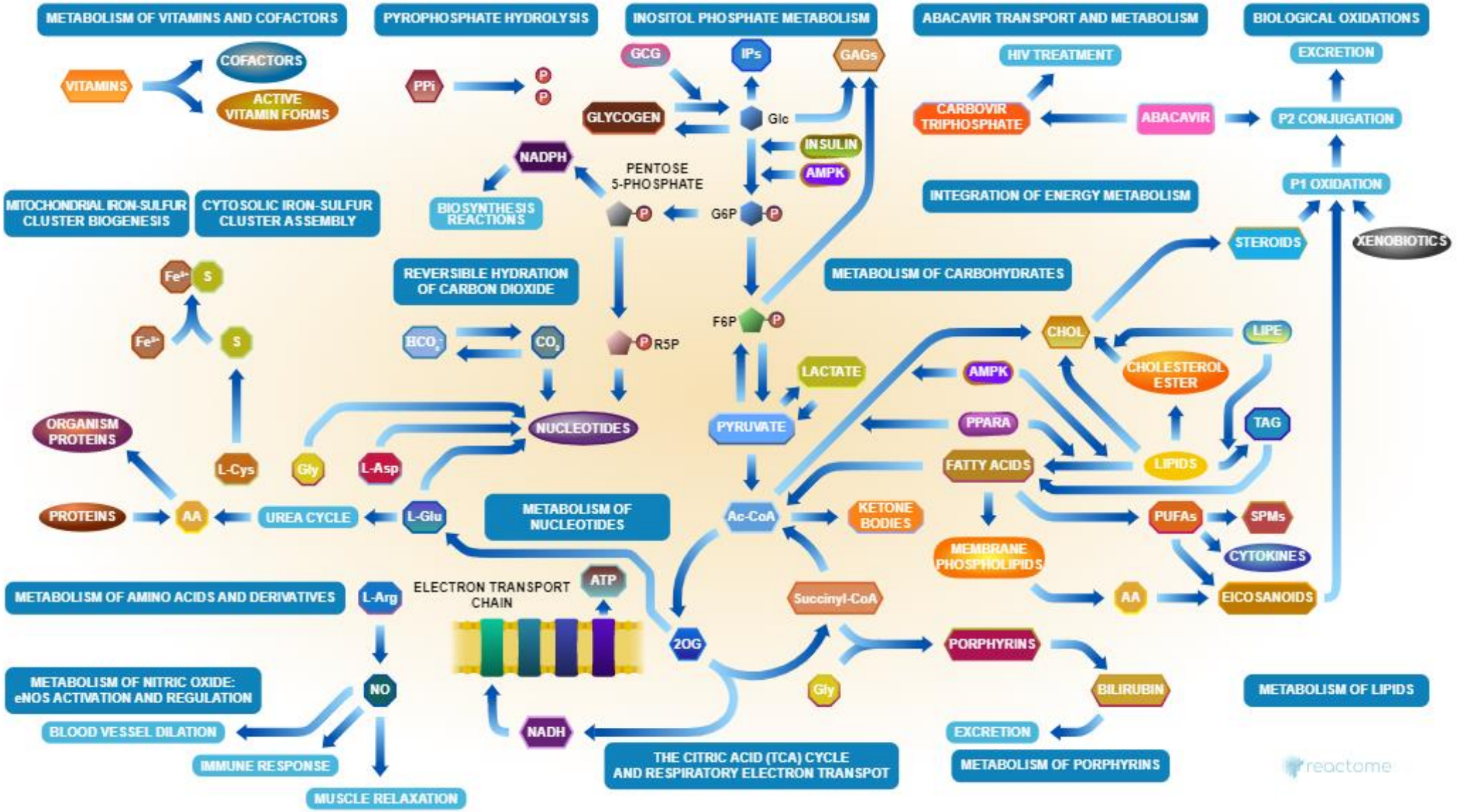
c.carbohydrates, lipids, proteins, and nucleic acids.

d.carbohydrates, lipids, and proteins.

What is Metabolism?

- Metabolism - the entire set of life-sustaining chemical reactions that occur in organisms.
  - The sum of all chemical processes involve in maintaining life.
  - These reactions number in the thousands and include reactions such as those responsible for getting energy from food, processing and removal of waste, building up muscles, growth, photosynthesis in plants, cell division, and reproduction.
  - Over the course of 40 years, while maintaining near constant weight, the human body processes about 6 tons of food and 10,000 gallons of water

What is Metabolism?

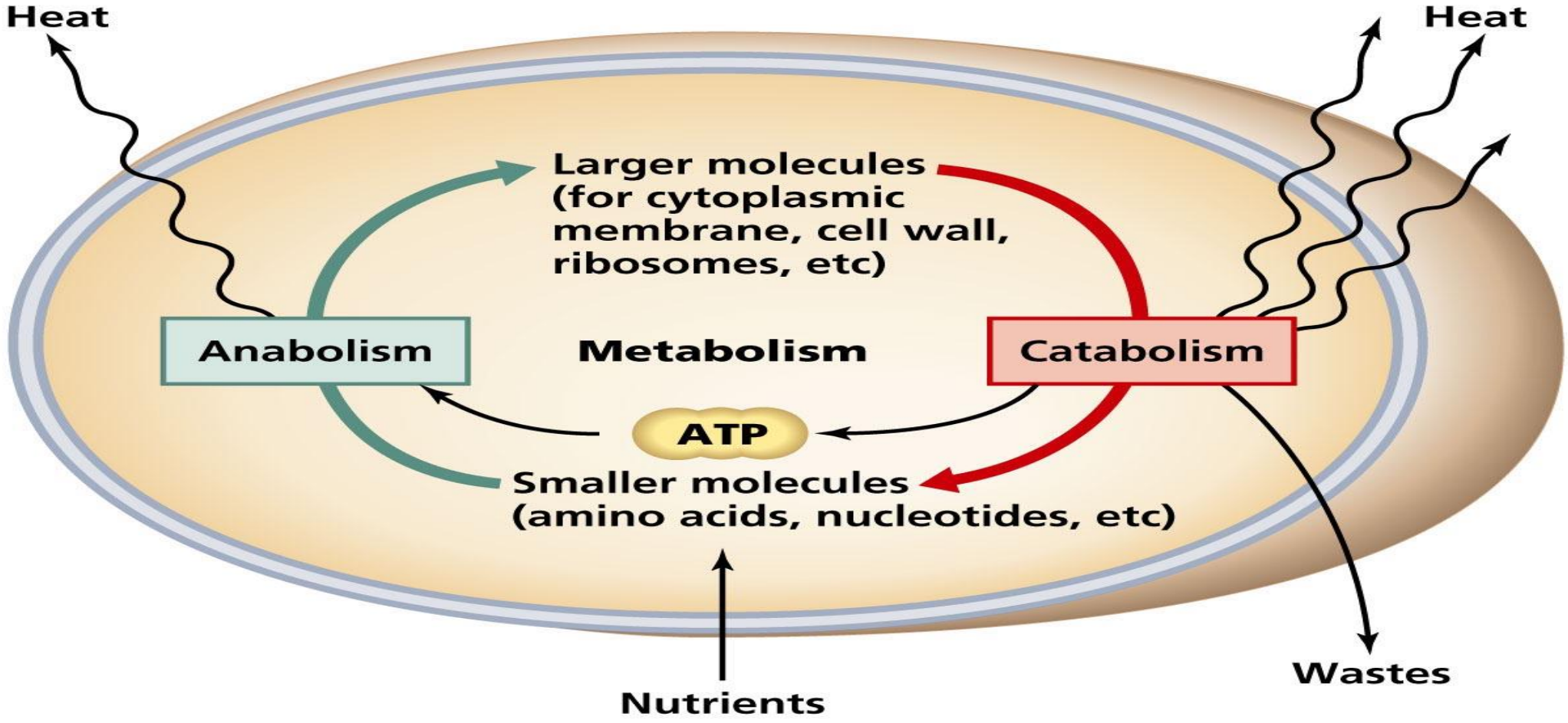


Overview

- The entire set of metabolic reactions is organized into smaller sets of sequential reactions called *metabolic pathways*.
- The species produced in the various reactions of a metabolic pathway are sometimes referred to as *metabolites.*
- Many of the reactions in metabolic pathways require enzymes; therefore, organisms can control (accelerate or suppress) metabolic pathways, according to their current needs, by upregulating, downregulating, inhibiting, or activating one or more of the enzymes involved in the pathway.

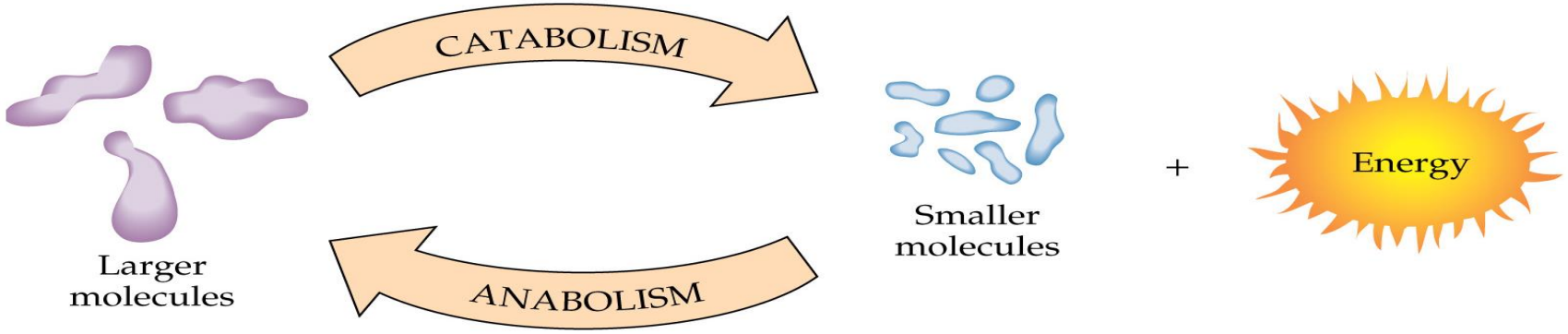
Types of Metabolism

Metabolic pathways can usually be classified as catabolic (catabolism) or anabolic (anabolism)



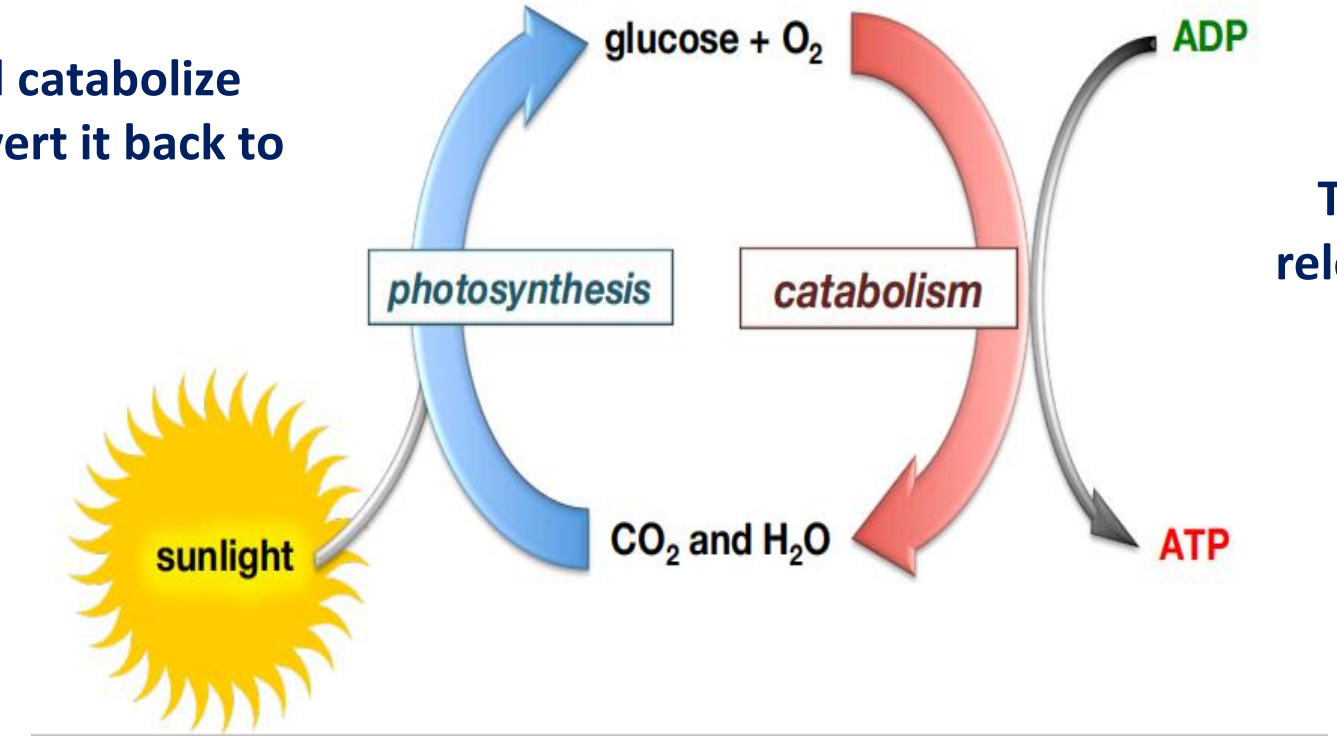
Types of Metabolism

- ***Anabolism*** *involves* ***building up*** *of smaller organic compounds to form larger ones and usually* ***requires energy***
  - *Example: protein synthesis, from nucleic acids to amino acids to protein*
- ***Catabolism*** *involves the* ***breakdown*** *of larger organic compounds into smaller compounds and usually* ***releases energy***
  - *Example: carbohydrates digestion, from starch to polysaccharide to maltose and dextrin to glucose*



Energy Production

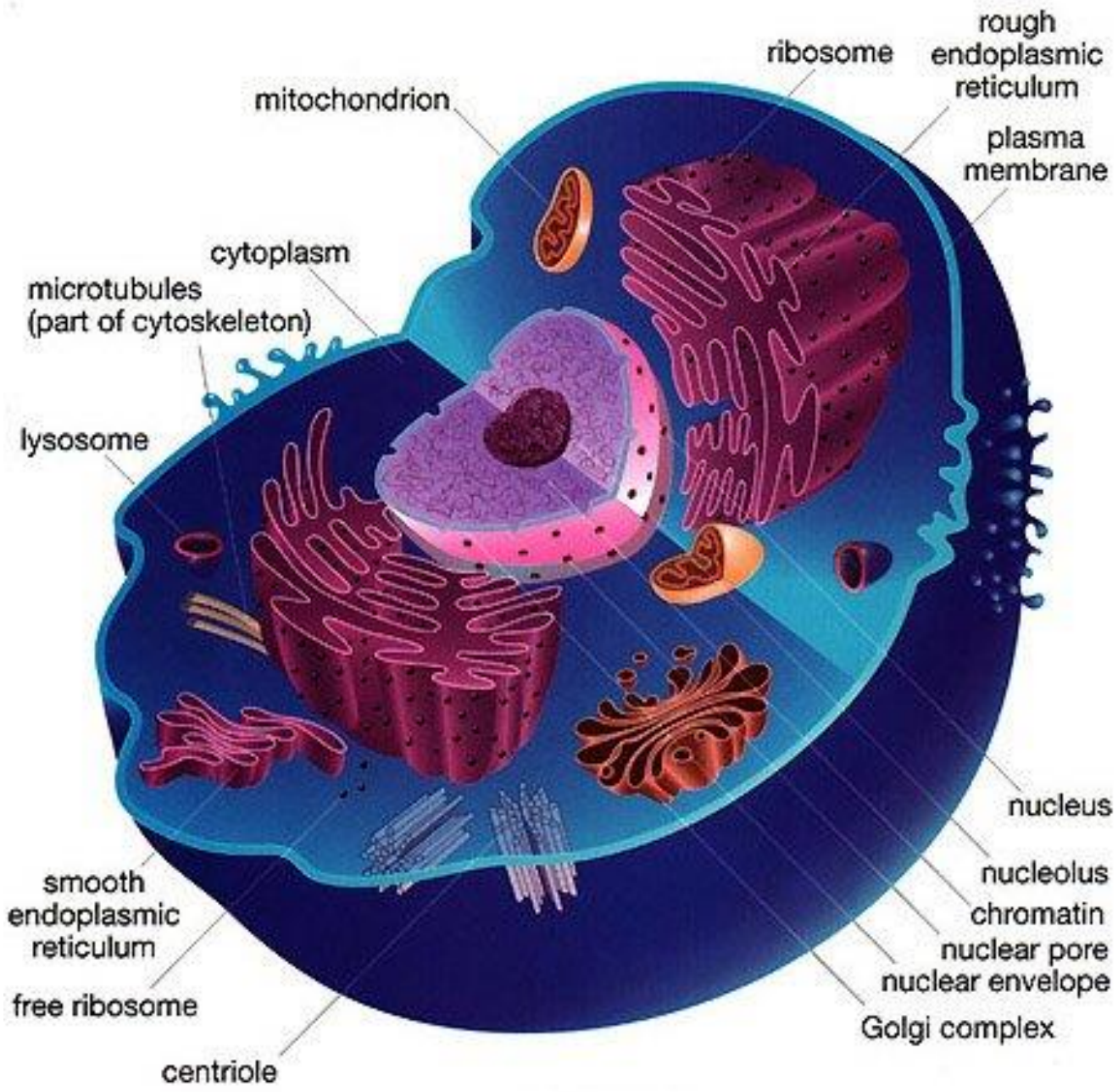
Organisms consume and catabolize macromolecules to convert it back to CO_2 and H_2O



These catabolic reactions release energy in the form of ATP

Sites of Metabolism

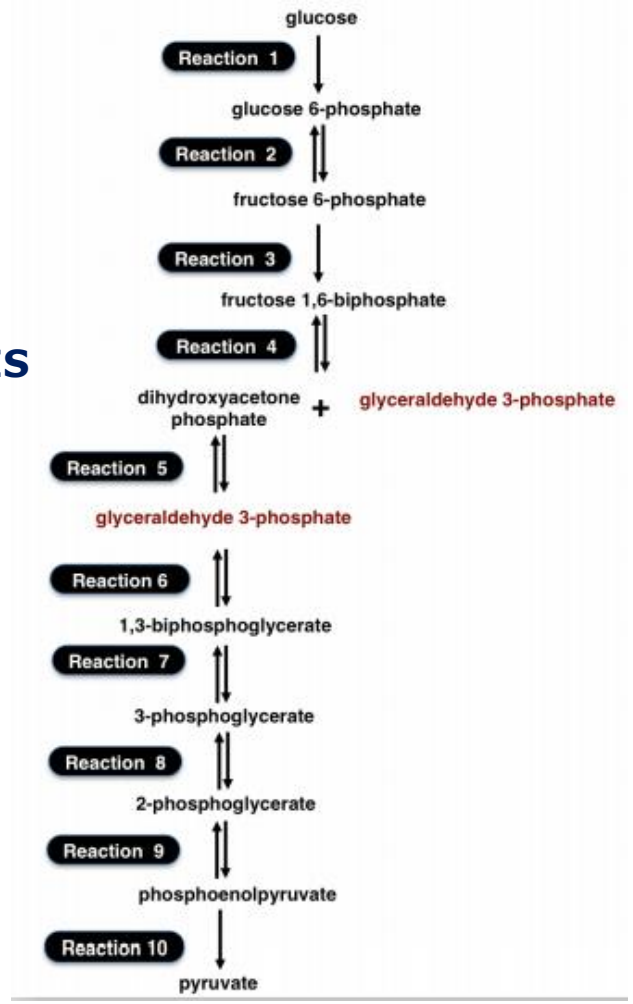
- ***Anaerobic Metabolism***
  - *no oxygen involved*
  - *Cytoplasm*
- ***Aerobic Metabolism***
  - *requires oxygen*
  - *mitochondria*



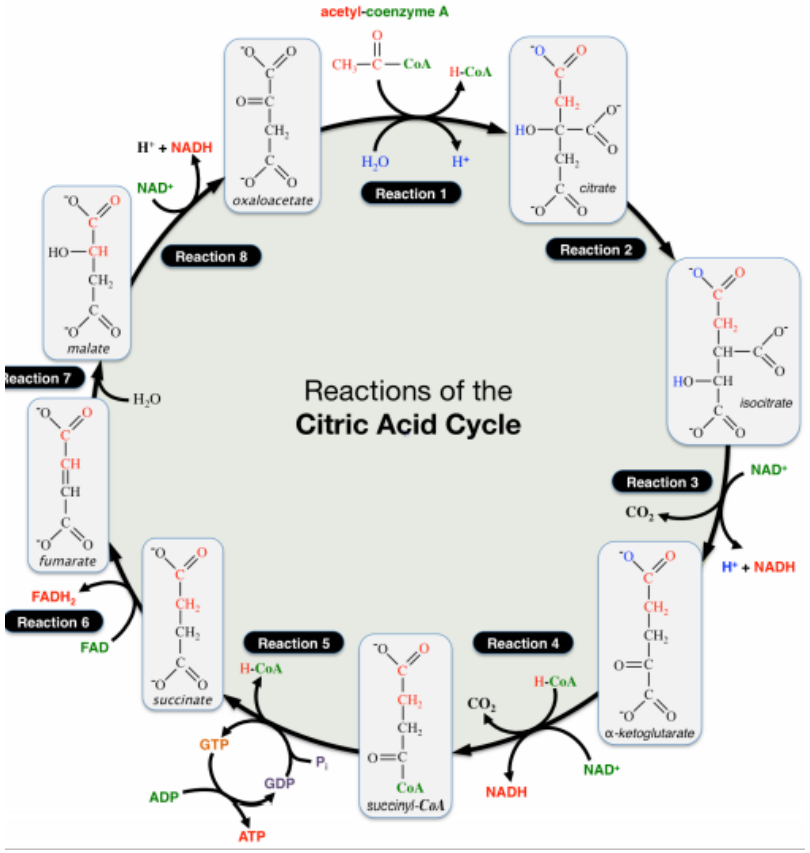
Types of Metabolic Pathways

- ***Linear*** *- A linear metabolic pathway is a series of reactions that are not repeated.*

GLYCOLYSIS



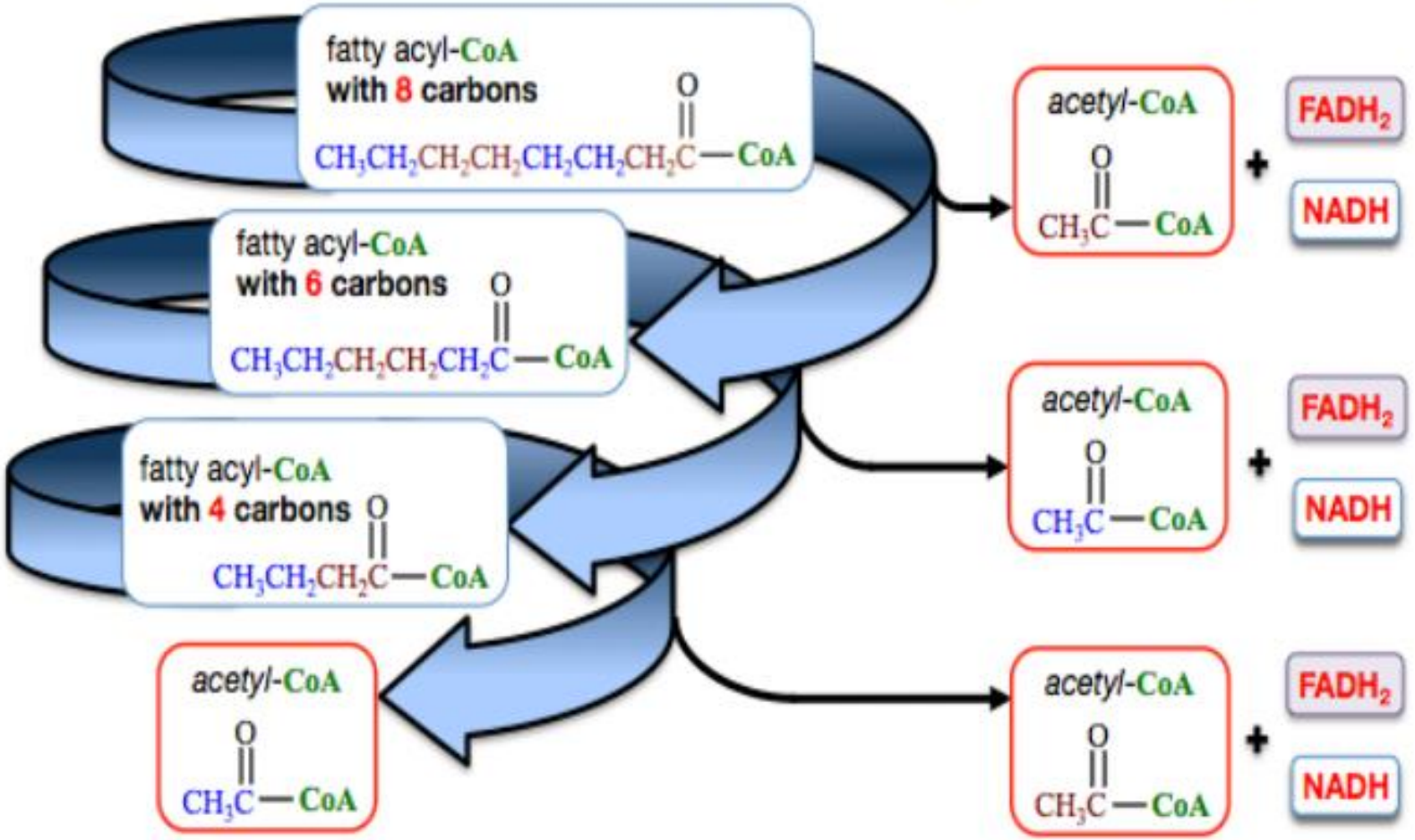
- ***Circular*** *- A circular pathway is a repeating series of reactions in which the final product is* ***also*** *an initial reactant.*



CITRIC ACID CYCLE

- ***Spiral*** *- A spiral pathway is a metabolic pathway in which a* ***series of repeated reactions*** *is used to break down (or build up) a compound.*

An *eight-carbon* fatty acyl-CoA will undergo ***three*** cycles of the ***β-oxidation*** reactions



Take note!

- *In this chapter, you will learn the series of chemical reactions that are involved in the metabolism of carbohydrates, proteins, and fats.*

- *The ultimate goal at the end of these reactions is to convert the chemical potential energy contained in food into chemical potential energy in the form of **ATP**.*

Exercise

Classify each of the following chemical processes as anabolic or catabolic.

Anabolic a. Synthesis of a protein from amino acids

Anabolic b. Formation of a triacylglycerol from glycerol and fatty acids

Catabolic c. Hydrolysis of a polysaccharide to monosaccharides

Copyright ©2016 Cengage Learning. All Rights Reserved

EXERCISE

What intermediate molecule in metabolic reactions is responsible for producing energy in the human body?

a. AMP

b. ADP

c. ATP

d. NADH

Copyright ©2016 Cengage Learning. All Rights Reserved.

REFERENCES

- Bettelheim, F., Brown, W., Campbell, M., Farrell, S., Torres, O.J., & Madsen, S. (2020) Introduction to General, Organic and Biochemistry (12th ed.). Cengage.
- Garrett, R.H. and Grisham, C.M. (2024). Biochemistry (7th ed.). Cengage Learning
- Stoker, S (2016). General, Organic, and Biological Chemistry 7th Ed., Cengage Learning.